

Wireshark is a **free and powerful packet analyzer** used to **capture and inspect network traffic** in real time. It's widely used by:

- **Network engineers**
- **Cybersecurity professionals**
- **Penetration testers**
- **Students** for learning networking



Main Uses of Wireshark

- Analyze **network issues** (latency, errors, slow connections)
- Detect **malicious traffic or attacks** (like DoS, ARP spoofing)
- Inspect **protocols** (HTTP, TCP, DNS, etc.)
- Reverse engineer malware or network behavior
- Monitor **live traffic** or analyze saved .pcap files



How Wireshark Works

- Captures packets from network interfaces (Ethernet, Wi-Fi)
- Displays detailed information: source/destination IP, ports, protocols, payload, flags
- Allows **filters** to narrow down what you see



Wireshark Display Filters (Very Useful)



Common Protocol Filters

Protocol	Filter Syntax
HTTP	http
DNS	dns
ARP	arp
TCP	tcp
UDP	udp
ICMP (Ping)	icmp



IP Address Filters

Task	Filter
Source IP	ip.src == 192.168.1.1
Destination IP	ip.dst == 192.168.1.5
Any IP Involved	ip.addr == 192.168.1.10



Port Filters

Task	Filter
------	--------

Source Port	tcp.srcport == 80
Destination Port	tcp.dstport == 443
Port (any direction)	tcp.port == 22 or udp.port == 53

Web & DNS Traffic

Protocol/Service	Filter
HTTP traffic	http
HTTPS (SSL/TLS)	ssl or tls
DNS requests/responses	dns

Error or Suspicious Traffic

Task	Filter
TCP Reset	tcp.flags.reset == 1
TCP SYN only (handshake start)	tcp.flags.syn == 1 and tcp.flags.ack == 0
Failed DNS resolution	dns.flags.rcode != 0

Example: Capture only HTTP traffic to a certain IP

```
plaintext
CopyEdit
http && ip.addr == 192.168.0.5
```

Capture Filters vs Display Filters

Type	Purpose	Example
Capture Filter	Before capture, limits what is captured	port 80
Display Filter	After capture, for analysis	http && ip.addr == x.x.x.x

Example Capture Filter

```
plaintext
CopyEdit
tcp port 443
Captures only HTTPS traffic.
```

Want to Practice?

You can download .pcap files from sites like:

- <https://www.malware-traffic-analysis.net/>
- <https://www.netresec.com/?page=PcapFiles>